Football(Soccer) Reference (FBref) Recommender System Analysis

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1) Selection of Recommender System

In this discussion, i chose to analyze the Football(Soccer) Reference platform **FBref.com** and specifically their "Player Comparison" and "Similar Players" engine. **FBref** is one of the most comprehensive soccer statistics websites, powered by **Opta data** from **Stats Perform**. Their recommender system identifies players with similar statistical profiles across various soccer leagues, which is a primary tool used by fans, scouts, and analysts to find "undervalued" talent or replacements for departing stars.

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2) Let's Perform the Scenario Design Analysis

As described below, Scenario design involves identifying the **three question framework**: "Who," "What ," and "Why or How" of a recommendation. In the context of building a recommender system for a soccer analysis platform like FBref, It is essential to perform the scenario design for both **the Organization** (FBref/Sports Reference) and **the User** (Fans/Scouts) is essential because their definitions of **"success"** are often fundamentally different and occasionally in conflict.

A. Organization Perspective (FBref)

- **Target User:** Advertisers and B2B data partners.
- **Goal:** Increase "stickiness" and page views.
- **Recommendation Value:** By showing similar players, the site encourages the user to click through multiple player profiles. This increases ad impressions and demonstrates the depth of their database to potential enterprise API customers.
- **Key Metric:** Average session duration and Click-Through Rate (CTR) on "Similar Players" lists.

B. User Perspective (The Customer)

- **Target User:** Soccer fans, fantasy managers, or professional scouts.
- **Goal:** Discover new players or validate scouting reports.
- **Scenario:** A user looks up *Jude Bellingham* and wants to find younger, cheaper alternatives for a Football Manager save or a scouting report.
- **Recommendation Value:** Saving time by filtering thousands of players down to the top 10 most statistically similar profiles.

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3. Reverse Engineering the System

Based on the interface and available technical documentation regarding their data partner (Opta), we can infer the following about FBref's recommendation engine:

1. **Content-Based Filtering (Statistical Profile):**
    - Unlike Netflix or Amazon, which use Collaborative Filtering (users who liked X also liked Y), FBref uses **Content-Based Filtering**.
    - The "features" are the player's percentile ranks in specific categories (e.g., Progressive Carries, Expected Assists (xA), Tackles, Aerial Wins).
    - They likely use a **Cosine Similarity** or **Euclidean Distance** algorithm on a normalized vector of stats to find the "nearest neighbors" in the high-dimensional data space.
2. **Segmentation by Position:**
    - The system first filters by "Position Group" (e.g., Fullbacks, Attacking Midfielders). It does not compare a Center Back to a Striker, regardless of how "efficient" their passing stats might both be.
3. **Data Normalization:**
    - Stats are analyzed "Per 90 Minutes" to ensure players with fewer appearances aren't unfairly penalized or inflated, though a minimum minute threshold is likely applied to maintain data integrity.

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4. Recommendations for Improvement

While FBref is industry-standard, its recommender system could be improved in the following ways:

I. Market Value/Age Weighting

- **Issue:** It often recommends players of similar quality who are equally expensive or older.
- **Improvement:** Add a "Smart Scouting" toggle that allows users to weight recommendations toward younger players or those in lower-tier leagues (e.g., "Find me players similar to 34 years old Kevin De Bruyne with a great passing and dribbling skills but under 23 years old in the Eredivisie").

II. Contextual/Tactical Filters

- **Issue:** The system is purely statistical and ignores team style.
- **Improvement:** Integrate **"Team Style"** similarity. In fact, a player might have similar stats to the Liverpool star right winger **Mohamed Salah**, but if they play in a low-block defensive system, they might not be a good recommendation for a high-pressing team.

III. Hybrid Collaborative Filtering

- **Issue:** It ignores user behavior.
- **Improvement:** Incorporate "Users also viewed" data. If professional scouts frequently jump from Player A's profile to Player B's, there may be a non-statistical link (e.g., similar physical build or injury history) that the data currently misses.

CONCLUSION

This analysis reveals that FBref's recommender system successfully bridges business objectives and user utility by leveraging position-specific, content-based filtering of Opta statistics. To maintain its competitive edge, the platform should evolve toward a hybrid model that incorporates market values and tactical context to provide more actionable scouting insights.